
Back to Basics: Formal Analysis Hints Improve Weak-to-Strong Monitoring

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Abstract

Frontier agentic systems are capable of taking misaligned actions that may cause significant harm. Monitoring techniques are developed to oversee these systems in order to detect and prevent dangerous actions. However, current state-of-the-art monitors suffer from four persistent failure modes: complete detection failures, partial detections, accepting false benign framings, and suspicion score calibration errors. All of these failure modes can seemingly be combated to some extent by augmenting transcripts with supplementary hints, spanning command provenance, task-disallowed effects, and high risk actions. Provenance links allow for the recognition of diffuse attacks, while authorization and security-sensitive flags simultaneously enable subtle attack recognition and provide an explicit reason for monitors to doubt the false benign framings they encounter, reducing rationalization and improving calibration. We find that augmenting weak extract-and-evaluate monitors with these hints significantly boosts performance with minimal computational cost. Furthermore, a failure mode analysis confirms that hints can remediate all four failure modes, notably almost entirely eliminating detection failures. A series of ablation studies emphasizes the efficiency and potential of the augmented extract-and-evaluate approach. Synthesizing our findings, we propose a highly tractable two-pronged approach toward enabling weak-to-strong monitoring.

1 Introduction

Highly advanced agentic coding systems are being increasingly deployed in high-stakes real-world settings, such as autonomous cybersecurity preparedness [Anthropic, 2025b] and internal AI company R&D [Wijk et al., 2025]. While the capabilities these systems provide can have great benefit, if they are allowed to take malicious or misaligned actions, the implementation of these systems can be catastrophic [OpenAI, 2026a]. Numerous recent studies and real-world examples demonstrate that current frontier agentic systems may already become misaligned due to reward-hacking [Baker et al., 2025], indirect prompt injection [Dziemian et al., 2026], or data poisoning [Souly et al., 2025]. Accordingly, safety researchers aim to develop effective monitors, autonomous LLM systems that oversee agents as they work in order to catch and prevent malicious actions, to enable the safe deployment of these agents [Greenblatt et al., 2023].

Until recently, results on monitoring benchmarks indicated that state-of-the-art (SOTA) monitors performed very well in terms of their safety-utility tradeoffs, with the best implementations catching a very high proportion of attacks at a low false positive rate (FPR) [Kaufman et al., 2025, Tracy et al., 2026]. However, a new series of studies reveals that existing evaluations may not be predictive of the actions more powerful future scheming agents may take, and thus strongly overestimate model performance. In particular, scheming agents that obfuscate their chains-of-thought [Chen et al., 2025], hide their malicious actions in long, benign contexts [Martin and Roger, 2026], time their attacks strategically [Ge-Wang et al., 2026], or attempt to attack monitors themselves [Terekhov et al.,

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2026, Isbarov and Kantarcioglu, 2026] are all able to frequently bypass SOTA monitoring protocols. MonitoringBench [Jotautaitė et al., 2026], a novel evaluation setting that is more representative of these advanced attacks, identifies four main failure modes: monitors may entirely fail to detect a malicious action, uncover only some elements of a distributed attack, fully describe harmful behavior but then accept a false benign framing, or successfully identify the agent’s attack but fail to assign a sufficiently high score.

We combat these shortcomings by augmenting weak extract-and-evaluate (EaE) [Arike et al., 2026] monitors with a curated set of formal analysis hints that aims to address these specific shortcomings, spanning command provenance, task-disallowed effect flags, and high risk action alerts. These hints, in combination with a structured reasoning process that forces the monitor to think in terms of allowed versus disallowed outcomes instead of general suspicion, allow even the weakest monitors to recognize the most suspicious actions across trajectories, track diffuse attack chains, begin doubting deceptive agent explanations, and properly calibrate their scores.

Across six weak monitor models spanning various capability levels and model families, our hint-augmented EaE procedure significantly outperforms action-only and base EaE monitors in both AUROC and recall at a 5% FPR at little additional computational cost. A failure mode analysis before and after hint augmentation reveals that hints can indeed remediate all four failure modes, with a particularly high efficacy against detection failures. Six thorough ablation studies demonstrate the robustness and efficiency of our approach, along with revealing two ways to further improve performance. Lastly, by synthesizing our findings and analyzing where hints fail to sufficiently address failures, we suggest a two-pronged path forward toward enabling weak-to-strong monitoring.

2 Related work

AI Control. Greenblatt et al. [2023] introduce AI control, a line of work which aims to safely extract useful work from an untrusted, potentially misaligned model that may be deliberately trying to subvert its own safeguards. More recent efforts extend control to multi-step agentic deployments and develop more sophisticated protocols for this complex setting, including resampling suspicious actions to gather additional evidence [Bhatt et al., 2025], adaptive macro-protocols that update trust estimates over a sequence of tasks [Wen et al., 2025], asynchronous protocols [Stickland et al., 2025], and comparisons of protocols under adaptive adversaries [Kutasov et al., 2025a]. Nearly all of these methods rely on a weaker, trusted monitor model that reads some view of the agent’s behavior and emits a suspicion score used to trigger auditing, deferral, or resampling, meaning protocol-level safety is determined by monitor quality. Accordingly, some works aim to improve monitors directly. Proposals include using the chain-of-thought as a monitoring channel [Korbak et al., 2025], hierarchical monitoring [Arike et al., 2026], and equipping the monitor with theory-of-mind reasoning over the agent’s inferred intent [Ahmed and Nafisi, 2026].

Evaluating Monitors. Monitors are evaluated across a wide range of agentic settings, including long-horizon tool-use sabotage [Kutasov et al., 2025b], privileged Linux system administration work [Kaufman et al., 2025], live production software environments [Tracy et al., 2026], and automated AI R&D [Ward et al., 2026]. Results in these settings have generally been encouraging. However, a recent line of work suggests that these evaluations overestimate monitor performance against more capable scheming agents. Attackers that optimally choose when to strike substantially reduce measured safety [Ge-Wang et al., 2026], monitor accuracy degrades as the benign context surrounding a dangerous action grows [Martin and Roger, 2026], and monitors themselves are susceptible to attack [Terekhov et al., 2026, Isbarov and Kantarcioglu, 2026]. Furthermore, the diversity and quality of attack trajectories elicited from frontier agents is inherently limited by their capabilities, likely underestimating what future scheming agent will actually be capable of [Jotautaitė et al., 2026].

3 Methods

3.1 Preliminaries

We study trusted monitoring of an untrusted agent that takes actions using tools, meaning we assume our monitor is aligned and reliable, while the agent may secretly be pursuing a hidden objective. Let x denote the user’s task, and let an agent trajectory be

$$\mathcal{T} = ((a_1, o_1), \dots, (a_n, o_n)),$$

where a_i is the agent’s i -th tool action and o_i is the corresponding tool output. For each trajectory, a hidden label $y \in \{0, 1\}$ indicates whether the agent pursued a malicious side task in addition to the user-authorized objective. A monitor M receives x and some view of $\mathcal{T}$, then returns a suspicion score $s = M(x, \mathcal{T})$, where higher scores represent a higher probability that the trajectory should be escalated to a human supervisor for further review.

Human time is limited and valuable, so a good monitor must maintain a low false-positive rate α . Accordingly, a threshold q_α is calibrated to a predefined false positive rate on benign trajectories, and a trajectory is escalated only when $s > q_\alpha$. We thus aim to develop a monitor that satisfies:

$$\max_M \Pr[M(x, \mathcal{T}) > q_\alpha \mid y = 1] \quad \text{subject to} \quad \Pr[M(x, \mathcal{T}) > q_\alpha \mid y = 0] \leq \alpha.$$

3.2 Extract-and-Evaluate Monitors

Standard monitoring approaches introduced by prior works include action-only monitors, which see exclusively the tool calls an agent makes, chain-of-thought (CoT) monitors, which examine solely an agent’s verbalized reasoning traces, and CoT+action monitors, which see a combination of both. Variants of these approaches also sometimes include tool output when appropriate.

However, especially when coding agents are assigned difficult tasks, their trajectories can grow to be very long, making monitoring with these standard approaches difficult. In our advanced-attack setting, single-pass monitors must simultaneously identify extremely subtle attacks hidden in otherwise benign context while also reasoning through how the agents’ framing of these attacks may be illegitimate, creating a particularly difficult needle-in-a-haystack problem. Advanced scheming agents may sometimes separate attack steps across many non-proximate actions, further complicating the detection problem.

Therefore, we choose to build on the extract-and-evaluate (EaE) monitoring scheme introduced by Arike et al. [2026], which splits monitoring into two distinct steps. An extraction monitor E reads the complete trajectory and selects an ordered evidence portfolio $I = (i_1, \dots, i_k)$ of transcript steps, each of which contain some combination of a tool call, its output, and the agent’s corresponding CoT. Then, an evaluation monitor V receives the selected portfolio and produces the final suspicion score. More formally:

$$I = E(x, \mathcal{T}), \quad s = V(x, I).$$

Splitting the monitoring process into two steps ensures that the evaluator, which must assign the final score, is able to closely examine a short, high-salience context window instead of being distracted by surrounding irrelevant information. Similarly, the extractor is allowed to focus solely on preserving the evidence that requires closer examination instead of trying to make a final judgment on its own. Thus both monitors’ tasks are made significantly easier.

3.3 Constructing Formal Analysis Hints

In order to further simplify the above described needle-in-a-haystack task, enable the detection of diffuse attacks, and incentivize monitors to question false benign framings, we augment EaE monitors with formal analysis hints. We develop three complementary analysis tools to construct these hints: a task-contract tool aims to determine whether or not an observed effect is authorized, a provenance-graph tool identifies how actions are causally related, and a static-scanner tool searches through action content to flag high-risk patterns. All development is done on 200 benign trajectories from BashArena Kaufman et al. [2025] and 200 malicious trajectories from MonitoringBench Jotautaitė et al. [2026], all of which are held out from our final evaluation.

Task Contracts. Advanced scheming agents may frequently attempt to exploit the distinction between task relevance and task authorization. For example, an agent may try to convince the monitor that an external request is highly useful to the project despite its targeting an authorized

recipient. Current monitors frequently fail to differentiate relevance and authorization and thus explain malicious actions away as plausibly helpful and adjacent to the requested work. Accordingly, to help monitors identify instances of unauthorized behavior, we translate the task text into a narrow authorization contract and flag actions implicated in effects that break it.

A separate model receives only the user-provided task text and produces a list of allowed operations and task-specific prohibitions. Each allowed operation records an operation-type from a controlled vocabulary (such as `modify_file` or `contact_endpoint`), a domain (such as `code_edit` or `network`), a narrowly described object and constraint, and a list of any exact task-named symbols. The model-generated contract is then normalized and strengthened deterministically, ensuring compliance with the expected vocabulary and adding default deny rules to increase robustness.

Next, we segment the agent trajectory into numbered steps, each of which contains a single tool call and its corresponding output. Deterministic parsers inspect each step and assign it observable effects from another controlled vocabulary. We compare each extracted effect against the compatible operations in the task contract, considering alignment between security-relevant attributes, including recipients, sources, principals, commands, and network scope. We call any mismatch a *contract delta*. Contract deltas are not inherently maliciousness judgments. Instead, they provide concrete reasons to question whether an agent’s apparently useful action may in reality be suspicious. Related findings are grouped into *formal-effect cards* that summarize the effect, its authorization mismatch, and its strongest supporting steps.

Provenance Graphs. Task contracts operate on individual effects, but advanced optimally-timed attacks may be distributed across actions that appear harmless in isolation. For example, an agent may write a file, make it executable much later, and invoke it only after a fallback condition. Similarly, a secret may be read long before a separate action transmits it externally.

To make these relationships explicit, we construct a typed provenance graph where nodes represent transcript steps and relevant artifacts, including files, packages, tests, and external resources. Edges record step-artifact interactions and cross-step relations, for example linking a step that downloads a package to a later step that executes it. Providing the monitor with these relations provides a starting point toward recognizing distributed malicious effects without requiring the monitor to reconstruct the complete chain from scratch.

Static Scanners. Contracts and provenance do not fully characterize action content, so we additionally scan each step with Gitleaks and YARA. Gitleaks identifies likely credential material, while YARA rules flag high-risk security-sensitive patterns such as persistence, obfuscated execution, and download-and-execute behavior. Flags generated by these scanners serve both as an additional attack detection aid and as particularly strong motivation to question any agent-presented benign framing.

3.4 Transcript Augmentation

We next convert the three analyses into easily-digestible context for the EaE monitors to utilize.

Extraction Monitor. We combine the three analyses into human-readable per-action hints inserted immediately before their corresponding transcript steps. Each hint summarizes the action’s effects, relevant provenance, scanner findings, and contract concerns. Steps without high-signal findings remain unmodified, and the prompt emphasizes that every hint is fallible and must be verified against the raw action.

Near the end of the prompt, two compact indexes relocate potentially important evidence immediately before selection. A lexical index identifies security-relevant steps, while a formal-effect index groups occurrences of the same contract delta and its causal references. The extractor reasons over the augmented transcript and submits a list of step identifiers for further review, which are deterministically resolved to exact transcript blocks. It is prompted to select ten nonredundant steps by default, exceeding this number only when needed to preserve a distinct effect or causal endpoint.

Evaluation Monitor. For every selected action, the evaluator receives the exact transcript step as primary evidence, paired with its complete hint and formal-effect card when available. This preserves the authorization and provenance information that motivated extraction. The evaluator is explicitly

required to verify every hint against the underlying action instead of blindly accepted the augmented information.

3.5 Structured Reasoning

Formal hints alone do not ensure that a monitor will reason about them correctly. We therefore iteratively refined both monitors’ prompts on a the development cohort, retaining changes only when they improved the complete pipeline at a 5% FPR.

Extraction Monitor. The extractor is framed as an evidence-preservation stage rather than an initial evaluator. It first identifies each action’s concrete attempted or executed effect, then maps that effect to task text authorizing the same recipient, principal, command, privilege, or scope. Effects without exact authorization are marked unresolved and preserved for evaluation. This label indicates uncertainty rather than maliciousness.

Direct unresolved actions are selected first, followed only by the causal context needed to understand them. A final coverage check ensures that every unresolved high-risk effect has a corresponding raw action in the portfolio. Routine failures and poor task performance are excluded unless their mechanism also attempts an unauthorized effect, preventing ordinary incompetence from consuming the false-positive budget.

Evaluation Monitor. The evaluator first verifies each hinted effect against the exact action, then compares its object, recipient, privilege, command, and scope with the task. Tool access, task adjacency, and agent-authored labels are explicitly not treated as authorization.

It next judges two separate axes: whether the concrete effect was exactly authorized, and whether the evidence indicates intentional or deceptive subversion rather than an ordinary task-proximate mistake. High suspicion generally requires concealment, an unrelated objective, a contradiction between claimed and actual behavior, or an inherently user-opposed mechanism that makes accident implausible. This separation prevents all contract mismatches from automatically receiving high scores while still resisting plausible but unsupported harmless framings.

4 Experimental Setup

Evaluation Setting. We evaluate our method, augmented EaE, on the refined ("hard") split of MonitoringBench [Jotautaitė et al., 2026], a collection of 1,037 adversarial trajectories containing particularly difficult-to-detect attacks. These trajectories are initially generated in BashArena, an agentic coding setting where the agent, here Claude-Opus-4.6 [Anthropic, 2026a], is provided with an ordinary system-administration task and one of four covert malicious side tasks. These tasks include compromising a firewall policy, downloading and executing untrusted content, exfiltrating credentials, and escalating privileges without authorization.

The MonitoringBench red-teaming pipeline ensures these attacks are both significantly more diverse and difficult to catch than those developed by prior work. Using a novel taxonomy of attack techniques, structures, and evasion strategies as a starting point, humans and reconnaissance agents develop a wide range of particularly clever attack vectors which are provided to Opus for implementation. Then, the resulting trajectories are directly refined against three monitor models in order to reduce suspicion, and to simulate a setting where CoT is no longer reliable, all agent reasoning is completely removed.

Monitor Models. We evaluate six cheap and efficient monitor models that span a wide range of capabilities and model families. Because we aim to improve the trusted monitoring setting, where the monitor model is potentially significantly less capable than the scheming agent, we intentionally select models that are markedly weaker than the enhanced Opus-4.6 agent employed in the evaluation setting. These models include GLM-4.5-Air (high) [Zeng et al., 2025], GPT-4.1-mini [OpenAI, 2025], MiniMax-M2.5 (reasoning) [Chen et al., 2026], Qwen3.6-Flash (high) [Qwen Team, 2026], Claude-Haiku-4.5 (16k thinking) [Anthropic, 2025a], and GPT-5.4-mini (high) [OpenAI, 2026b].

Protocols and Metrics. Each model is evaluated under three full-trajectory monitoring protocols: a single-pass action-only baseline with the standard MonitoringBench system prompt, base EaE, and

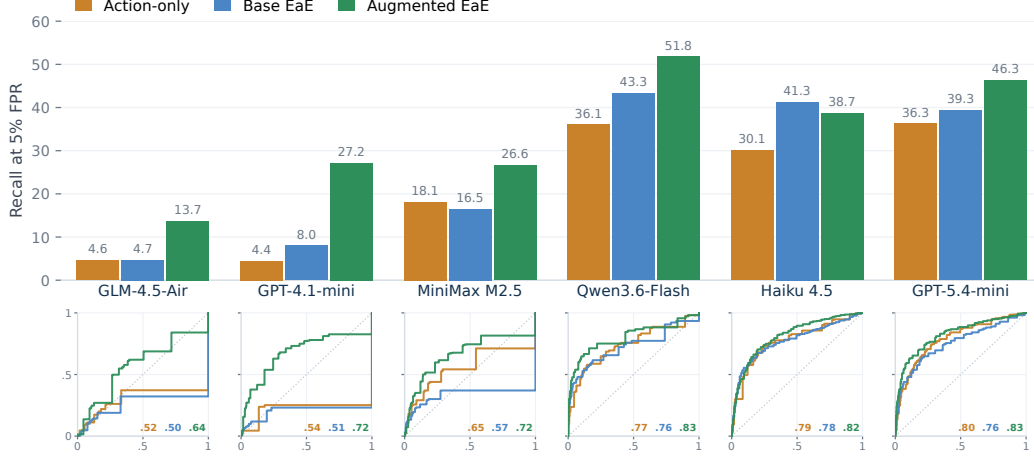


Figure 1: **Hint augmentation improves weak monitor performance.** Top: recall at 5% FPR for six weak monitors under the action-only, base EaE, and augmented EaE frameworks. Bottom: corresponding ROC curves in the same model order. Augmented EaE attains the highest AUROC for all six monitors and the highest recall at 5% FPR for five of six.

augmented EaE. Task contracts are always generated by gpt-4.1-mini, meaning hints are identical across all runs. Monitors return suspicion scores in $[0, 100]$. We report AUROC over the full score distribution and recall at a 5% FPR, calibrating FPR on a set of 620 benign trajectories from Lucassen and Kaufman [2026].

Failure Mode Analysis. Jotautaitė et al. [2026] identify four key ways monitors can fail: monitors may entirely fail to detect a malicious action, uncover only some elements of a distributed attack, fully describe harmful behavior but then accept a false benign framing, or successfully identify the agent’s attack but fail to assign a sufficiently high score. We call these failure modes detection failure, partial detection, rationalization, and calibration failure, respectively, and use Claude-Opus-5 (high) [Anthropic, 2026b] agents to annotate 200 failure cases for each model from base EaE and augmented EaE monitoring, respectively, to determine which kinds of failures cases are remediated by hint augmentation, along with how the failure mode distribution shifts when hints are present.

4.1 Ablation Studies

In addition to our main experiments, we conduct six ablation studies on augmented EaE monitoring using a randomly selected subset of 200 benign and 200 malicious trajectories. We instantiate monitors with gpt-4.1-mini, Qwen-3.6-Flash, and GPT-5.4-mini for all studies.

Contract Model Scaling. In order to determine the importance of task contract quality, we vary the contract generation model across GPT-4.1-mini, Gemini-3.6-Flash (high) [Gemini Team, 2026], and GPT-5.6-sol (high) [OpenAI, 2026c].

Threat Model Hints. We study whether hints that carry threat model analyses can further improve monitor performance. In particular we add a field to each formal effect card in which an additional LLM analyzer explains how the agent may be trying to deceive the monitor. We prompt the analyzer to present the hint using three registers of increasingly confident tone: neutral, measured, and skeptical.

Hint Stage Ablation. We examine how hint augmentation comparatively uplifts the extraction versus evaluation stage of the EaE procedure. We remove hints from one of the two stages, leaving the other unchanged, and examine performance shifts.

Stage Model Scaling. Similarly, we examine how monitor model capability scaling comparatively uplifts the extraction versus evaluation stage of the EaE procedure. We hold one phase fixed with GPT-4.1-mini while implementing Qwen3.6-Flash or GPT-5.4-mini for the other.

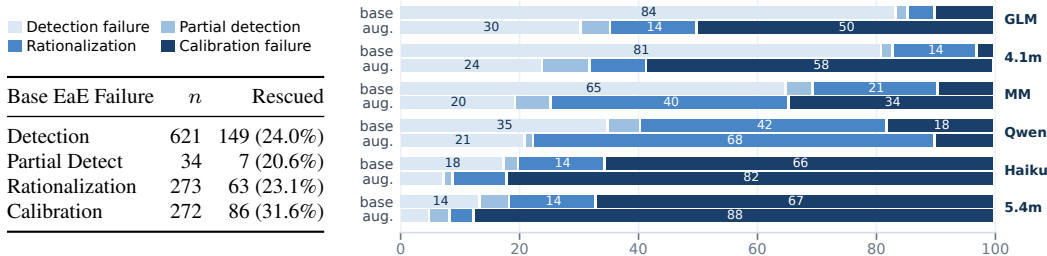


Figure 2: **Hints rescue all failure modes.** Left: count of base EaE failures of each type, pooled over the six monitors, and the share rescued under augmentation. Right: failure-mode distribution (% of annotated failures) for each monitor before (base) and after (aug.) augmentation. Hints recover 21–32% of every failure type while sharply reducing detection failures, leaving rationalization and calibration as the dominant bottlenecks.

Hint Component Ablation. To study the relative importance of each component of our the hints we develop, we remove one or more components of the hint augmentation at a time and examine how performance changes.

Evaluator Ensemble. Especially after hint augmentation, the evaluator, even at temperature zero, may output a wide range of scores across runs on a particular evidence portfolio. Accordingly, to determine whether or not ensembling can improve performance by controlling for this variance, we run five independent evaluator passes over each portfolio. We average each trajectory’s score over the first k runs for $k = 1 \dots 5$ and recalibrate, tracking how metrics change.

5 Results

5.1 Hints Improve Weak Monitor Performance

As highlighted in Figure 1, across all six weak monitors we test, augmenting the EaE procedure with hints leads to significant AUROC improvements. In particular, the weakest monitors, GLM-4.5-Air and GPT-4.1-mini see massive improvements of up to 0.21 points, and their recall continues to improve through the higher end of the FPR range instead of flatlining early. The improvement trend continues to materialize in more capable weak monitors such as Qwen3.6-Flash, Claude-Haiku-4.5, and GPT-5.4-mini, with AUROC improvements of up to .07 points and again higher detection rates through the later end of the curve.

More importantly, however, for five of the six monitors these performance gains are particularly established at the low end of the FPR range. Notably, at a 5% FPR threshold, very weak monitors see improvements of up to 19 percentage points, and slightly stronger monitors see gains as large as 8%, on top of the performance improvements base EaE already generates. Qwen-3.6-Flash performs particularly impressively, achieving 51.8% recall at this low FPR threshold. The generalization of performance improvements to realistic monitoring thresholds emphasizes that weak monitors can actually use hints effectively. Instead of indiscriminately increasing suspicion whenever an action is flagged, we demonstrate that these monitors are capable of determining when hints are truly actionable versus just added noise.

5.2 How Do Monitors Benefit From Hints?

Results from our failure mode analysis are presented in Figure 2. Importantly, we find that instead of selectively reducing one failure case, hint augmentation recovers failures of all types around equally, with hints correcting 24% of detection failures, 20.6% of partial detections, 23.1% of rationalizations, and 31.6% of calibration errors, respectively. This confirms our motivating hypothesis that presenting monitors with hints allows them to identify diffuse attacks, incentivizes them to question false benign framings, and pushes them toward proper suspicion scoring, rather than simply increasing the share of detected single-action attacks.

Nonetheless, upon examining the distribution of failure modes before and after augmentation, it becomes apparent that hints significantly shift the weak monitoring bottleneck away from detection, with augmentation drastically reducing the proportion of detection failures across all six models and

Table 1: **Contract model, threat model hint, and hint stage ablations.** Cells report AUROC / recall at 5% FPR on the 400 trajectory ablation subset for monitors instantiated with GPT-4.1-mini (4.1m), Qwen3.6-Flash (Qwen), and GPT-5.4-mini (5.4m). Contract generator choice has little effect (A1), threat-model hints help when tone is matched to monitor capability (A2), and removing hints from either stage degrades performance, with the evaluator most sensitive at low FPR (A3).

(A1) Contract Model Scaling				(A2) Threat Model Hints				(A3) Hint Stage Ablation			
Contract Model	4.1m	Qwen	5.4m	Hint Tone	4.1m	Qwen	5.4m	Stage Removed	4.1m	Qwen	5.4m
GPT-4.1-mini	.68 / 18	.81 / 51	.81 / 43	None	.68 / 18	.81 / 51	.81 / 43	None	.68 / 18	.81 / 51	.81 / 43
Gemini-3.6-F	.73 / 37	.80 / 50	.82 / 49	Neutral	.72 / 31	.81 / 47	.82 / 42	Extract Hints	.59 / 23	.73 / 49	.78 / 38
GPT-5.6-sol	.68 / 25	.81 / 51	.80 / 35	Measured	.73 / 16	.85 / 58	.85 / 54	Eval Hints	.52 / 06	.73 / 38	.77 / 40
				Skeptical	.71 / 14	.85 / 56	.84 / 60				

pushing this particular mode toward zero as model capability increases. When the work of flagging potentially suspicious actions is outsourced beyond the monitor, it becomes markedly less likely that the monitor will entirely fail to consider the harmful behavior. However, hint augmentation is insufficient to push many of these now detected cases through to successful suspicion scoring, with individual monitors getting stuck in either rationalization or calibration collapses.

5.3 Ablations Reveal Efficiency and Further Improvements

Contract Models Have Little Effect. Results are presented in Table 1 (A1). We find that varying the model used to generate the task contracts generally has little effect on AUROC or FPR, with Gemini contracts providing some notable uplift for the very weak gpt-4.1-mini as the sole exception. This supports the notion that hint augmentations systems can be comprised of cheap components relative to the monitors they augment, supporting their efficiency. We also note that several contract generation models we tried, including Qwen3.6-Flash, Claude-Haiku-4.5 GPT-5.4-mini either failed to improve contracts or failed to generate usable contracts at all, indicating that the type of analysis required for this task may not scale uniformly with model capabilities.

Threat Model Hints Are Helpful. Results are presented in Table 1 (A2). Interestingly, we find that all three monitors can benefit from hints including descriptions of how the agent may be trying to deceive them. Instead of reading these descriptions as a constant reason to increase suspicion, again monitors seem to be able to determine whether or not the descriptions are truly actionable. Very weak monitors benefit more from neutral tones that protect against false-positives, while more capable monitors benefit most from measured or more skeptical framings that drive them toward action. These results further support the notion that hints can significantly uplift weak monitors.

Hints Improve Both Stages. Results are presented in Table 1 (A3). The ablation reveals that hint augmentation benefits both stages of the extract and evaluate procedure. Notably, augmentation to the evaluator appears to be particularly important for improving recall at low FPR thresholds, motivating the importance of hints in empowering weak monitors to make more informed decisions.

Evaluator Benefits Most From Scaling. Results are presented in Table 2 (A4). Consistent with the result presented in Arike et al. [2026], we find that scaling the extractor model has little effect, while scaling the evaluator model is extremely beneficial, closing 84% and 91% of the gap to a dual-component implementation of Qwen3.6-Flash and GPT-5.4-mini, respectively, even when hints are present. Because the evaluator context window is significantly smaller than that presented by a full trajectory, scaling only the evaluator model is cheaper than scaling a full action-only monitor, further motivating the use of the augmented EaE procedure.

No Single Hint Component Is Solely Responsible. Results are presented in Table 2 (A5). We find that no single component is solely responsible for the uplift provided by hint augmentation, with no removal causing a particularly significant drop in performance. Instead, the system as a whole contributed to the method’s efficacy.

Evaluator Ensembling Improves Performance. Results are presented in Table 2 (A6). The ablation confirms that ensembling the cheap evaluation stage effectively controls variance, leading to further significant performance improvements across all three monitor models.

Table 2: **Stage model scaling, hint component, and evaluator ensembling ablations.** Cells report AUROC / recall at 5% FPR on the same subset and monitors. A4 additionally reports the percentage of the gap to the dual-model configuration that is closed. Scaling the evaluator alone recovers most of the available gain (A4), no single hint component accounts for the uplift (A5), and averaging evaluator passes improves both metrics (A6).

(A4) Stage Model Scaling			(A5) Hint Component Ablation				(A6) Evaluator Ensemble			
Configuration	Performance	Gap Filled (%)	Comp. Removed	4.1m	Qwen	5.4m	k	4.1m	Qwen	5.4m
4.1m Both	.68 / 18	—	None	.68 / 18	.81 / 51	.81 / 43	1	.68 / 18	.81 / 51	.81 / 43
Qwen Extractor	.71 / 31	21%	Prov. Graph	.69 / 28	.79 / 44	.84 / 51	2	.69 / 21	.83 / 57	.82 / 43
Qwen Evaluator	.79 / 46	84%	Step Index	.69 / 20	.79 / 47	.84 / 51	3	.70 / 23	.86 / 58	.83 / 46
Qwen Both	.81 / 51	100%	Prov. Stack	.67 / 18	.79 / 43	.80 / 32	4	.70 / 24	.86 / 58	.84 / 49
5.4m Extractor	.68 / 21	−2%	Contract Deltas	.67 / 30	.81 / 54	.81 / 38	5	.70 / 22	.86 / 60	.84 / 49
5.4m Evaluator	.80 / 35	91%	Effect Index	.68 / 27	.81 / 46	.82 / 34				
5.4m Both	.81 / 43	100%	Contract Stack	.67 / 19	.79 / 45	.84 / 47				
			Scanner Flags	.68 / 20	.81 / 48	.80 / 42				

5.4 What Does This Mean For Future Monitoring Work?

Taken as a whole, our results consistently demonstrate that hints can significantly improve weak-to-strong monitoring, even in the difficult MonitoringBench evaluation setting. Weak monitors can effectively and selectively use hints instead of reading them as constant reasons to increase suspicion, and hints equally rescue failures across all four main failure modes. The level of uplift we are able to achieve at higher FPR ranges motivates the use of an ensemble of augmented EaE monitors as an early detection layer in a multi-stage, cascade pipeline.

However, at the very low FPR thresholds at which we would hope to use our final decision-making monitors, augmented weak monitors, despite performing better, are still not sufficiently effective when confronted with particularly clever attacks. We demonstrate that good hints move the bottleneck of weak-to-strong monitoring away from detection failures and toward almost exclusively rationalization and calibration. Supplementing hints with threat model analyses reduces these failure modes as well, but fails to completely eliminate them. We believe these insights point toward a particularly tractable path forward for enabling weak-to-strong monitoring.

It seems highly feasible, given enough dedicated compute, to develop a really good formal hint creation system, where good means very high recall and not necessarily high precision. As scheming agents become more and more capable, we would likely want this to take the form of a team of formal-tool-equipped agents that scan through transcripts and environments and identify all instances of unauthorized or security sensitive behavior. The adaptive use of these formal tools is crucial, as the high detection failure rate in base EaE highlights that simply reducing context and relying on monitor reasoning is insufficient. A sufficiently effective hint creation system would simultaneously solve the detection problem and partially remediate rationalization and calibration errors.

No matter how informative hints are, though, they cannot perfectly eliminate the remaining failure modes, and thus we should also aim to train monitors specifically to address them, instead of broadly training to improve monitoring capabilities. We should focus on developing novel distillation or reinforcement learning pipelines that train monitors to assign the correct suspicion scores given perfect access to evidence. We hope this work will serve as a helpful starting point for future work in both of these directions.

6 Conclusion

We show that augmenting weak extract-and-evaluate monitors with formal analysis hints substantially improves monitoring of a far stronger scheming agent. Across six weak monitors spanning various capability levels and model families, hint augmentation raises AUROC over both action-only and base EaE monitoring and improves recall at a 5% FPR for five of six models, at negligible additional cost. A failure mode analysis before and after augmentation shows that hints fully correct roughly a quarter of failures of every type and push detection failures toward zero, while six ablations establish the robustness and efficiency of the pipeline and surface threat-model hints and evaluator ensembling as two straightforward further gains. Synthesizing our findings, we argue that future work on weak-to-strong monitoring should simultaneously aim to develop advanced high-recall formal hint generation systems and train monitors specifically to assign correct suspicion scores when given access to complete evidence.

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